

MASTER'S COMPREHENSIVE EXAMINATION

Part I Theory (90 minutes) December 1st, 2012

Instructions: Attempt any three out of the following four questions on both pages. Mention which three you would like to have graded. Open book and open notes.

1. A physicist measures the length of an object using two methods, and let X_A and X_B be the measurements. Assume X_A and X_B are jointly normal, and $X_A \sim N(\mu, 3^2)$, $X_B \sim N(\mu, 4^2)$, where μ is the true length of the object. Assume also that X_A and X_B are correlated with $\text{Corr}(X_A, X_B) = 0.2$.

- If the physicist uses the weighted average $X = 0.7X_A + 0.3X_B$ as the final measurement value, what is the distribution of X ?
- If the physicist used the formula $X = \delta X_A + (1 - \delta)X_B$ as the final measurement value, what is the distribution of X ?

2. A theater sells tickets for three seating areas, A, B, C, with 20, 30, 50 seats respectively. If all seats are occupied for a show, but ticket holders are assigned random empty seats upon arrival. Find the expected number of ticket holders sitting in the correct area indicated on their tickets.

3. Let X_1, X_2 be independent gamma variables with parameters $\alpha = 2, \beta$, i.e.

$$f_X(x; 2, \beta) = \frac{e^{-x/\beta} x}{\beta^2}, x > 0.$$

- (a) Find a sufficient statistic.
- (b) Find the MLE of β .

4. Let X be a single observation from a population whose density is

$$f_X(x; \theta) = \theta x^{\theta-1}, \quad 0 < x < 1 \\ 0, \quad \text{otherwise.}$$

Find the most powerful test of size $\alpha = .05$ for $H_0: \theta=1$ vs $H_1: \theta=2$.